

Date: 2024-11-11

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Subject: UL Solutions project number 4791347598 E541626

Dear Tong Shen,

I have completed your project plan and this letter outlines important project information about the UL Solutions service(s) you requested.

UL Solutions has received enough information to start the product evaluation; however additional items and/or samples may be needed to complete the evaluation. Details regarding the investigation items and/or samples required are included below under the heading of Required Investigation Items and Required Samples. Providing the items and samples requested below will avoid unnecessary delays in your project.

Product service description:

Order Number 15336831 - Power Transformer, SF-XXXXX/YY.YYY series, HV: 36 KV~48 KV, Capacity: 750~16667kVA IEEE C57.12.00 & C88, XQNU/7

Standard(s) referenced for this evaluation:

IEEE C57.12.00, STANDARD GENERAL REQUIREMENTS FOR LIQUID-IMMERSED DISTRIBUTION, POWER, AND REGULATING TRANSFORMERS, Issue Date 11/09/2021
IEEE C57.12.90, TEST CODE FOR LIQUID-IMMERSED DISTRIBUTION, POWER AND REGULATING TRANSFORMERS - CORRIGENDUM 1: EDITORIAL AND TECHNICAL CORRECTIONS, Issue Date 11/09/2021
CSA C88, Power Transformers and Reactors, Edition 5, Issue Date 11/2016

Product category:

XQNU/7

Required investigation item(s) for evaluation:

BOM, Sample, drawing

Required sample(s) for evaluation:

Samples (Rec'd/Req'd)	Sample Description
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Please print the sample shipping form(s) attached to the email and use as the shipping label or include it in each package when shipping the required samples to UL Solutions. This is required to associate your samples to this project.

Assumption(s):

- <xxxxxxx>
- All safety critical components, enclosures, and materials are UL Listed/Recognized and suitable for the intended application.
- All safety critical components, enclosures, and materials are UL and c-UL Listed/Recognized and are suitable for the intended application (for UL and c-UL evaluations).
- Quotation does not include fees associated with travel, lodging, meals, shipping, etc.
- Quotation does not include fees associated with Safety Certificates for other countries.
- No alternate components are employed.

Proposed test plan:

Test items	Standard	Clause
Ratio, Polarity and phase relation tests	IEEE C57.12.90	6, 7
	IEEE C57.12.00	9.1, table 17
	CSA C88-16	16.2 b), c)
Resistance Measurements	IEEE C57.12.90	5
	IEEE C57.12.00	table 17
	CSA C88-16	16.2 a)
Insulation resistance	IEEE C57.12.90	10.11
	IEEE C57.12.00	Table 17
	CSA C88-16	16.2 O), 16.6
Lightning impulse	IEEE C57.12.90	10.3
	IEEE C57.12.00	5.10.7.1, table 5, table 17
	CSA C88-16	16.2 i), 16.3.4
Applied-voltage tests	IEEE C57.12.90	10.6

	IEEE C57.12.00	table 17, Clause 5.10.5.2
	CSA C88-16	16.2 g), 16.3.2
Induced-voltage test	IEEE C57.12.90	10.7
	IEEE C57.12.00	table 17, Clause 5.10.5.3
	CSA C88-16	16.2 h), 16.3.3
No-load losses and excitation current at 100% & 110%	IEEE C57.12.90	8
	IEEE C57.12.00	table 17, 5.9, 9.3
	CSA C88-16	16.2 d), k), 16.4.1, 16.4.2
Measurement of short-circuit impedance and load loss	IEEE C57.12.90	9
	IEEE C57.12.00	table 17, 5.9, 9.3
	CSA C88-16	16.2 e), 16.4.3, 16.4.4
Zero-phase-sequence impedance	IEEE C57.12.90	9.5
	CSA C88-16	16.2 r)
Audible sound levels	IEEE C57.12.90	13
	IEEE C57.12.00	table 17, 8.2.3
	CSA C88-16	16.2 q), 16.8.1
Temperature Rise	IEEE C57.12.90	11
	IEEE C57.12.00	table 17, 5.11
	CSA C88-16	16.2 f)
Dissolved gas in oil analysis	IEEE C57.12.00	Table 17
	CSA C88-16	16.5

Test items	Standard	Clause
Partial discharge test	IEEE C57.12.00	Table 17
Control (auxiliary) and cooling consumption losses	IEEE C57.12.00	Table 17
Pressure and Leak	IEEE C57.12.00	Table 17
	CSA C88-16	16.2 i), 16.9
Low-frequency test on auxiliary devices and control	IEEE C57.12.00	Table 17
	CSA C88-16	16.2 n)
Operation tests of all devices	IEEE C57.12.00	Table 17
Mechanical tests	CSA C88-16	16.5
Lifting and moving devices	IEEE C57.12.00	Table 17
Insulation power factor and capacitance	IEEE C57.12.00	10.10
	CSA C88-16	20.4.6 b)

Timeline assessment:

- Customer requested completion date: <xxxxxx>
- Estimated project completion date: 2025-03-10

Any proposed project plan and timeline included in this communication is based on the received information and on the above assumptions. The test program and other aspects of the project plan will be finalized following a comprehensive construction review of your specific product.

A project handler from our engineering staff will begin the investigation shortly. The project handler will keep you informed of the progress of your project. If you have any questions or need additional information, you can contact me or customer service.

Sincerely,

Sam Wu
Project Engineer
Sam.X.Wu@ul.com

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